

Next, use this script for automatic updates and management of the sheet using the API key. Follow the deployment instructions (see Fig. 3) as outlined in the previous DIY article.

Software configuration

In the Arduino IDE, install the library named TINY GPS and the software serial library for ESP 8266. In the code, import the necessary libraries and configure the Wi-Fi. Insert the Google Sheet API key into the code. Configure the software serial pin to which you have connected the GPS module; you can use any available pin on the board.

Now, create the setup function and initialise the serial and software serial with the appropriate baud rate supported by your GPS. In our design, we are using the AIR530 GPS with a default baud rate of 9600, so we set the same baud rate in the setup function.

Next, create a loop to check the GPS data and upload it to the Google Sheet. In our prototype, we use Wi-Fi for internet connection and data uploading. However, in actual IOT logistics tracking, they may use direct-to-satellite-based IOT chips for sea-based cargo or GSM-based tracking systems. Regardless of the system, the process remains

the same: connect the device to the internet and upload real-time data to the sheet.

Fig. 4 provides a code snippet, while Fig. 5 illustrates capturing and sending GPS data to the Google Sheet.

Construction and testing

After assembling the circuit as shown in Fig. 2, power the device and wait a few seconds. It will start uploading location data to the Google Sheet, along with the cargo/shipment ID numbers, as displayed in Google Sheets. Fig. 6 showcases the real-time location and destination data for different cargos, along with maps.

In Google Sheets, you can customise functions and add more data as needed. You can include sections for distance left to cover, parcels that are stuck, extended location inactivity, or rerouting shipments. Customise the location and map views according to your preferences and maintain all shipments and logistics in a central sheet. This allows you to access graphs and analytics for all of them.

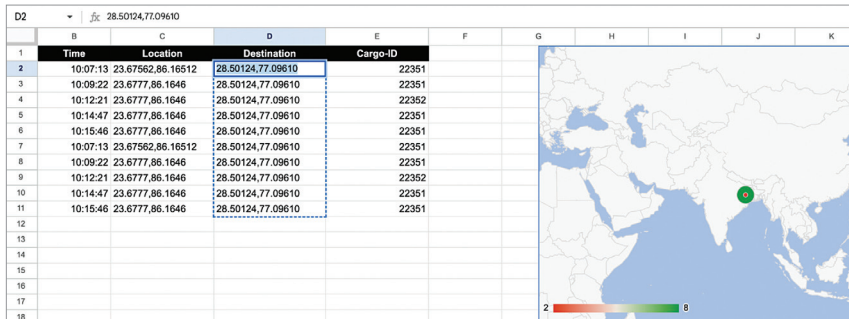


Fig. 6: Sheet showing the location and destination data in real time of different cargo along with the map

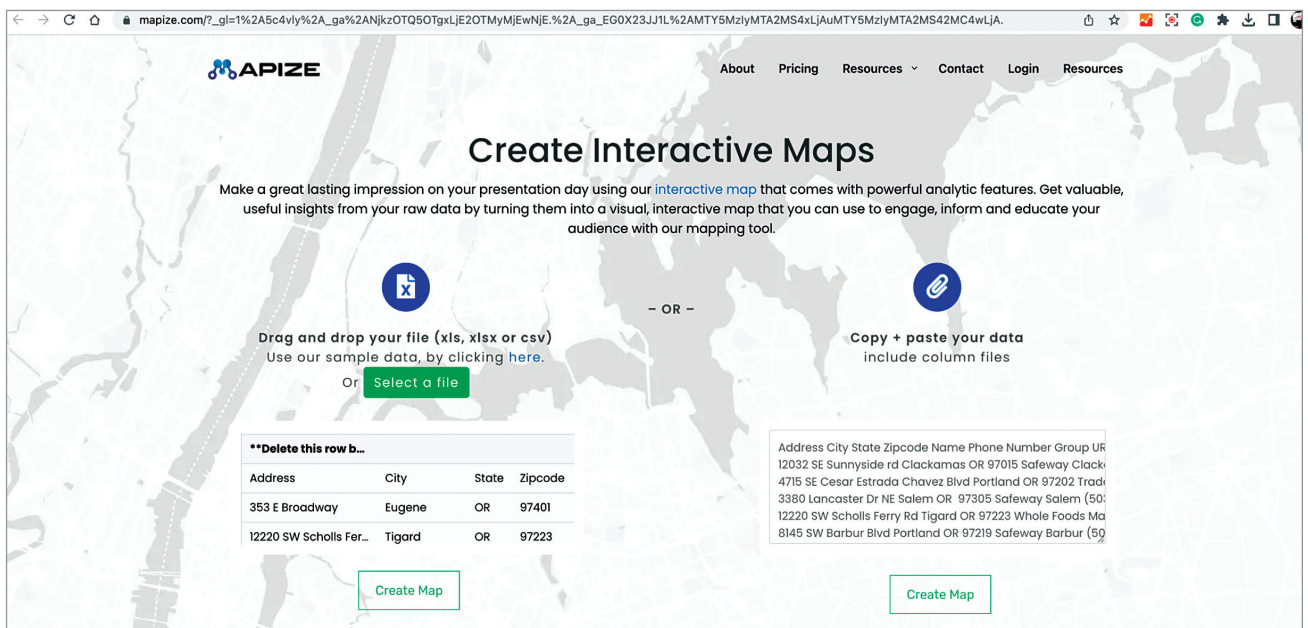


Fig. 7: Route and map making website